

# Discovering genuine resting state brain networks via simultaneous component analysis

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# Outline of Talk

## 1) Resting State Data

- Task versus Rest
- rsEEG Data

## 2) Finding Brain Networks

- Bilinear Components
- Simultaneous Components

## 3) Single Subject Example

- Local vs Global Changes
- Results from Subject One

## 4) Multi-Subject Example

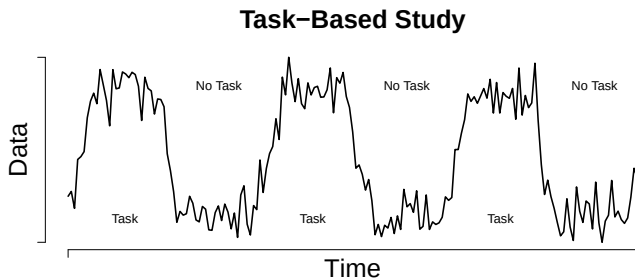
- Intra- vs Inter-Subject Variance
- Results from All Subjects

# Resting State Data

# Task-Based Studies

**Task-based** studies collect brain imaging data while a subject performs some experimental “task” of interest.

- Classic block design or event-related design
- Expect subjects to have some measurable response to task

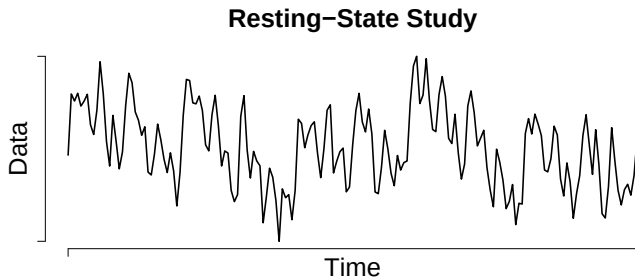


**Figure 1:** Depiction of hypothetical data from a task-based study.

# Resting State Studies

**Resting-state** studies collect brain imaging data while a subject “rests” with eyes closed or open.

- No event/task to temporally synchronize responses
- Expect to discover underlying brain organization (networks)



**Figure 2:** Depiction of hypothetical data from a resting-state study.

# Open Source rsEEG Data\*

Data from 12 healthy subjects

- All right-handed
- Males and females

Collected at 5 kHz from 62 channels

- Reference Cz
- Ground below Oz

Using 5 min (300 sec) of data

- 1,500,000 time points per channel per subject

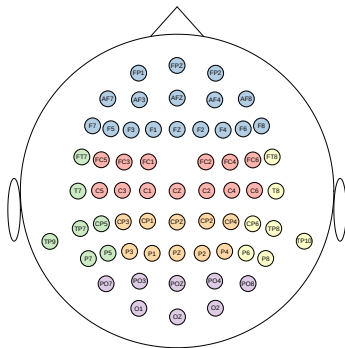
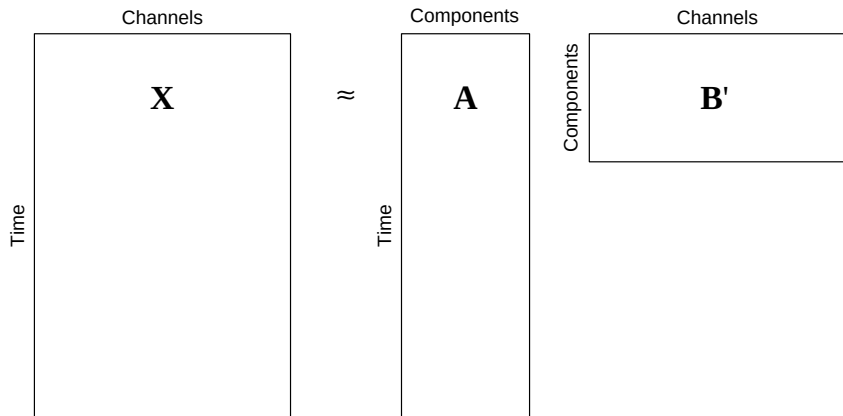


Figure 3: The 62-channel EEG cap.

\*From Sockeel, Schwartz, Pélégriani-Issac, and Benali (2016)

# Finding Brain Networks

# Bilinear Model Visualization



**Figure 4:** Depiction of the bilinear model, which includes FA, PCA, and ICA.

# Bilinear Model Definition

Let  $\mathbf{X} = \{x_{ij}\}_{I \times J}$  with  $x_{ij}$  denoting the data at time point  $i$  from channel  $j$ .

A **bilinear model** assumes that

$$x_{ij} = \sum_{r=1}^R a_{ir}b_{jr} + e_{ij} \quad \text{or} \quad \mathbf{X} = \mathbf{A}\mathbf{B}' + \mathbf{E}$$

where

- $\mathbf{A} = \{a_{ir}\}_{I \times R}$  with  $a_{ir}$  denoting the score of time point  $i$  on component  $r$
- $\mathbf{B} = \{b_{jr}\}_{J \times R}$  with  $b_{jr}$  denoting the loading of channel  $j$  on component  $r$
- $\mathbf{E} = \{e_{ij}\}_{I \times J}$  with  $e_{ij}$  denoting the error at time point  $i$  and channel  $j$

# Rotational Indeterminacy Problem

Bilinear models cannot uniquely find networks because

$$\mathbf{AB}' = \mathbf{ATT}^{-1}\mathbf{B}' = \tilde{\mathbf{A}}\tilde{\mathbf{B}}'$$

where  $\tilde{\mathbf{A}} = \mathbf{AT}$  and  $\tilde{\mathbf{B}} = \mathbf{B}(\mathbf{T}')^{-1}$  for any nonsingular matrix  $\mathbf{T}$ .

Problem: different rotations change our interpretation of components.

Need some assumptions for a unique decomposition:

- PCA assumes orthogonality for  $\mathbf{A}$  and  $\mathbf{B}$
- ICA assumes independence for  $\mathbf{A}$  or  $\mathbf{B}$

# Rotational Indeterminacy Visualization

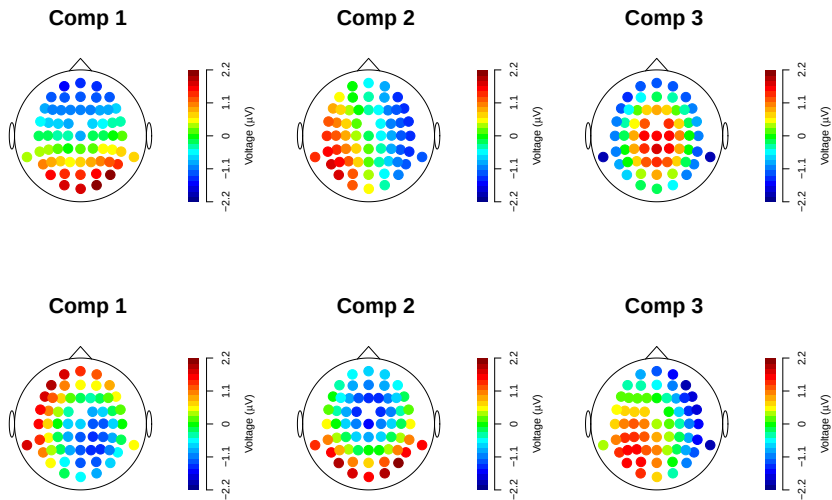


Figure 5: Depiction of two orthogonal rotations of  $R = 3$  components.

# Cattell's (1944) Principle of Parallel Proportional Profiles

Raymond Cattell argued that genuine factors (or components) should vary proportionally depending on their salience during particular occasions.

Cattell proposed to rotate to “simultaneous simple structure” and suggested this offered a unique solution to the rotational indeterminacy problem.

However, Cattell never fully formulated a factor/component model to solve the rotational indeterminacy...

# Cattell's Insight into Inter-Relations Between Factor Models

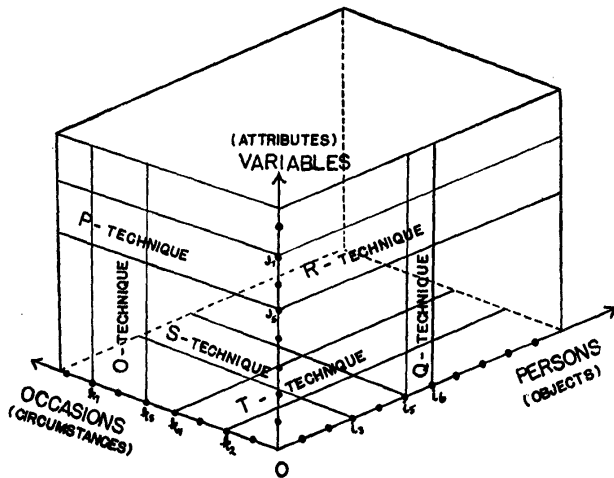


Figure 6: “The Covariation Chart” from Cattell (1952).

# Harshman's (1970) Parallel Factors (Parafac) Model

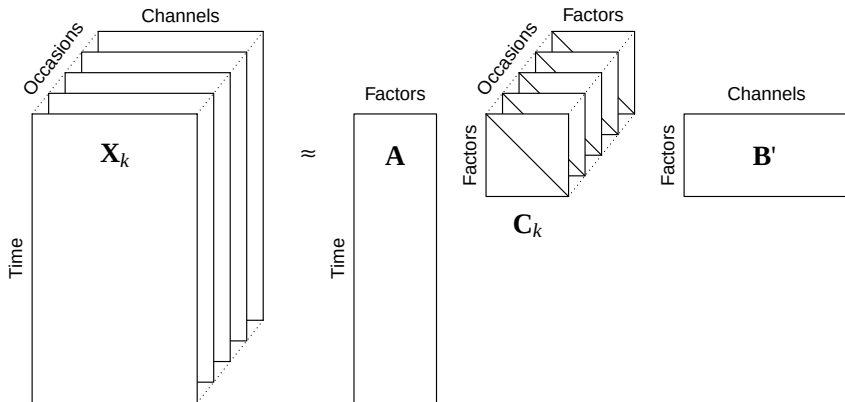


Figure 7: Depiction of the Parallel Factors (Parafac) model.

# Potential of Parafac for Task-Based Studies

Letting  $\mathbf{X}_k = \{x_{ij(k)}\}_{I \times J}$  denote the  $I$  time points by  $J$  channels matrix from the  $k$ -th occasion, the Parafac model assumes

$$\mathbf{X}_k = \mathbf{A}\mathbf{C}_k\mathbf{B}' + \mathbf{E}_k$$

where

- $\mathbf{A}$  and  $\mathbf{B}$  have same interpretation as bilinear model
- $\mathbf{C}_k = \text{diag}(c_{k1}, \dots, c_{kR})$  are the factor loadings for the  $k$ -th occasion
- $\mathbf{E}_k = \{e_{ij(k)}\}_{I \times J}$  is the error matrix for the  $k$ -th occasion

Parafac can produce essentially unique (rotationally invariant) solutions.

- **Intrinsic axis property** of trilinear model

# Harshman's (1972) Parallel Factors-2 (Parafac2) Model

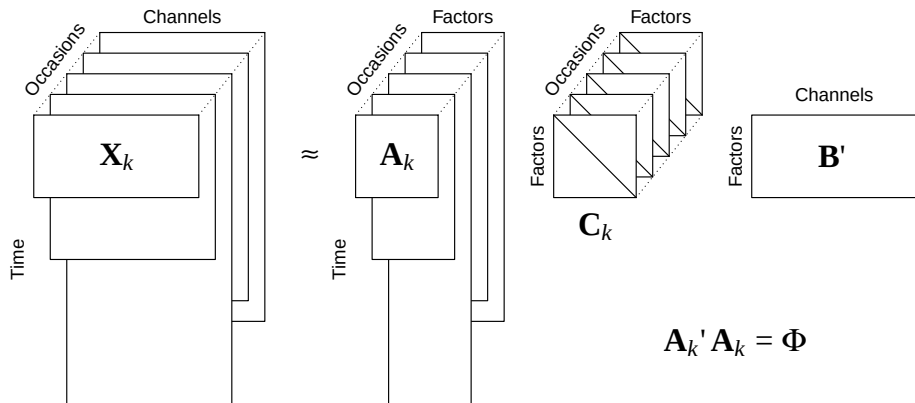


Figure 8: Depiction of the Parallel Factors-2 (Parafac2) model.

# Potential of Parafac2 for Resting-State Studies

Letting  $\mathbf{X}_k = \{x_{ij(k)}\}_{I_k \times J}$  denote the  $I_k$  time points by  $J$  channels matrix from the  $k$ -th occasion, the Parafac2 model assumes

$$\mathbf{X}_k = \mathbf{A}_k \mathbf{C}_k \mathbf{B}' + \mathbf{E}_k \quad \text{subject to} \quad \mathbf{A}_k' \mathbf{A}_k = \mathbf{\Phi}$$

where

- $\mathbf{A}_k = \{a_{ir(k)}\}_{I_k \times R}$  with  $a_{ir(k)}$  denoting the score of time point  $i$  on factor  $r$  specific to occasion  $k$
- $\mathbf{\Phi} = \mathbf{A}_k' \mathbf{A}_k$  is the common crossproduct matrix
- Other terms have same interpretation as Parafac model

Parafac2 can also produce essentially unique (rotationally invariant) solutions.

# Simultaneous Component Analysis Framework

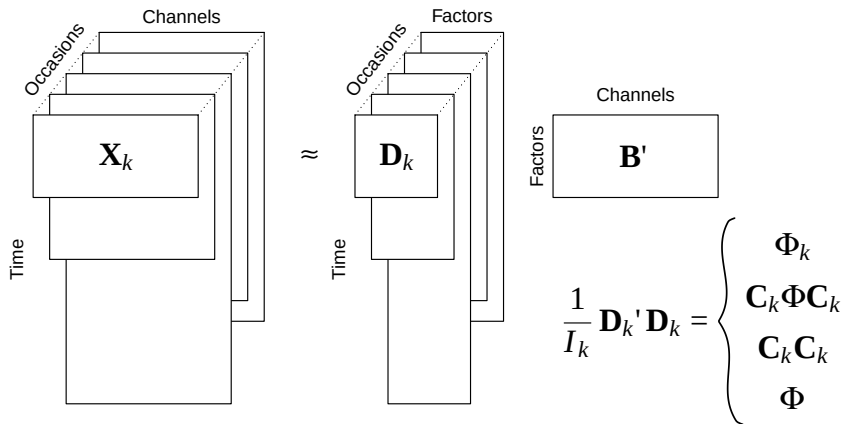


Figure 9: Depiction of Timmerman and Kiers's (2003) SCA framework.

# Single Subject Example

# Bandpass Filter Data

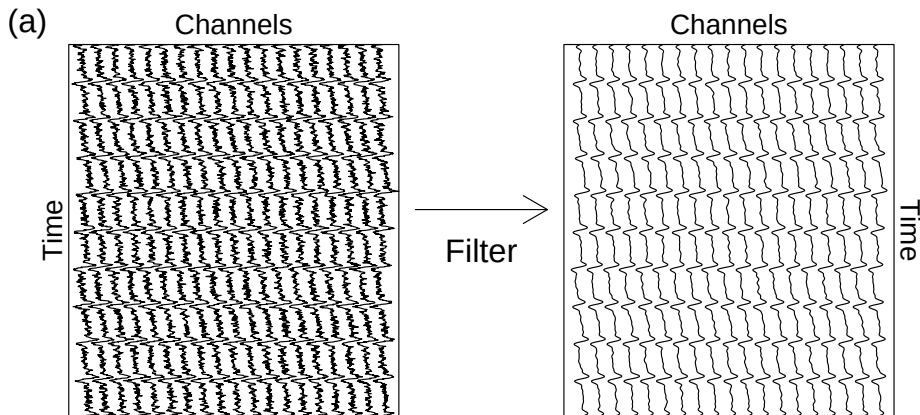


Figure 10: Bandpass filter each channel using some reasonable range (e.g., 1–50 Hz).

# Slice and Center Data

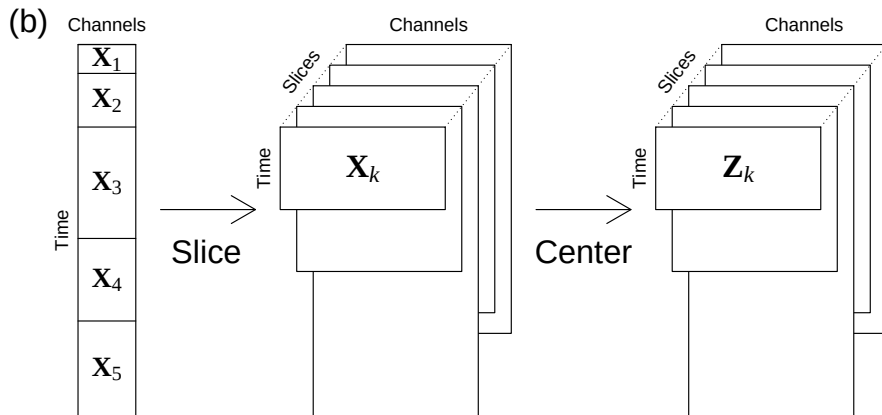


Figure 11: Slice and center to form multiple occasions.

# Simultaneous Components (Parallel Factors) across Slices

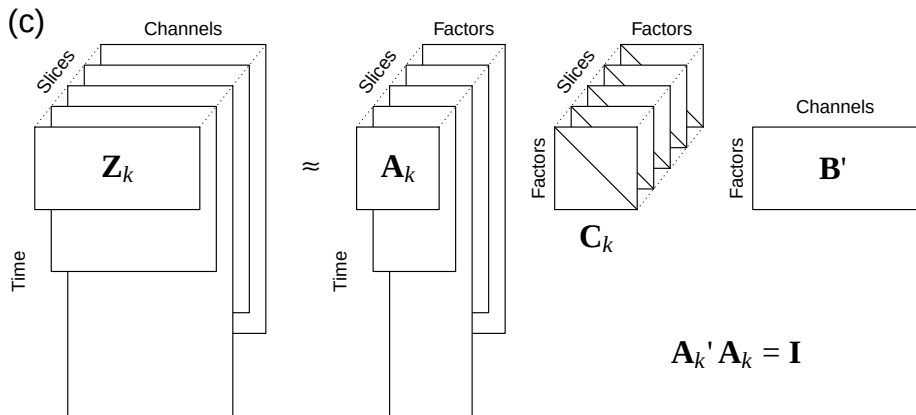


Figure 12: Parafac2 with orthogonal Mode A weights (SCA-IND).

# Choosing Number of Factors

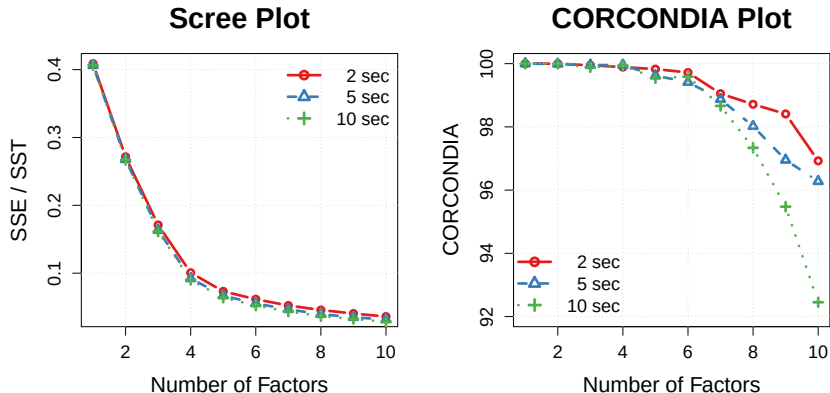


Figure 13: Scree and core consistency diagnostic (CORCONDIA) plots.

# Factor Loadings for Mode B for Subject 1

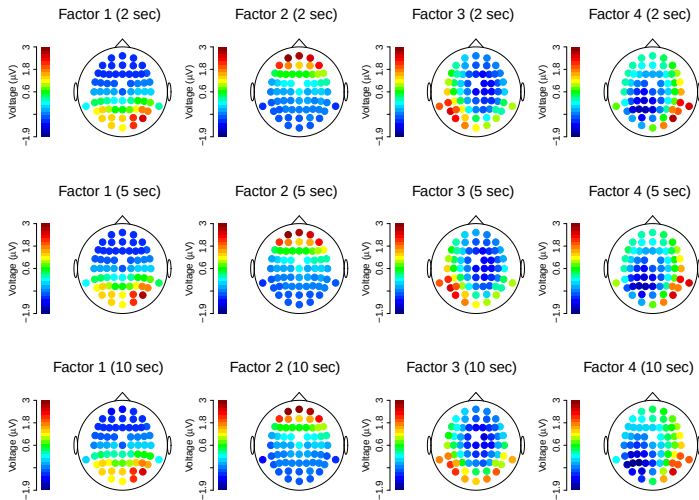


Figure 14: Estimated loadings for Mode B (channels).

# Power Spectral Density of Factor Scores for Subject 1

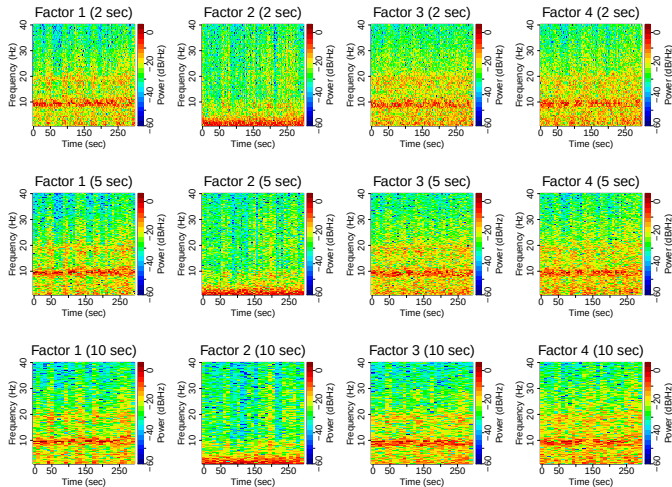


Figure 15: Estimated power spectral density of Mode A (time) weights.

# Factor Loadings for Mode C for Subject 1

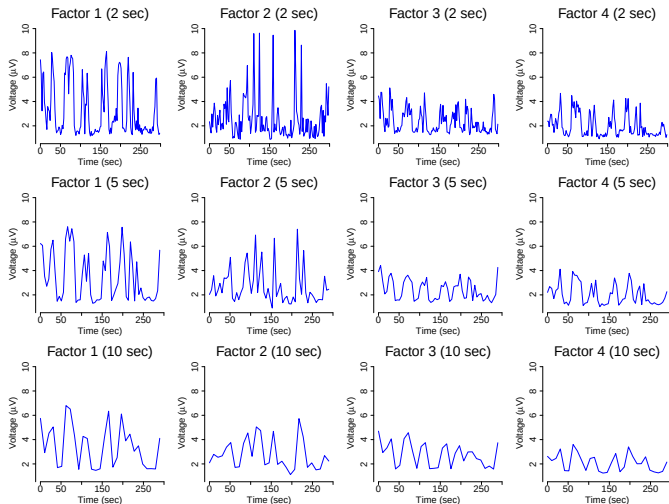


Figure 16: Estimated loadings for Mode C (slices).

# Power Spectral Density of Data for Subject 1

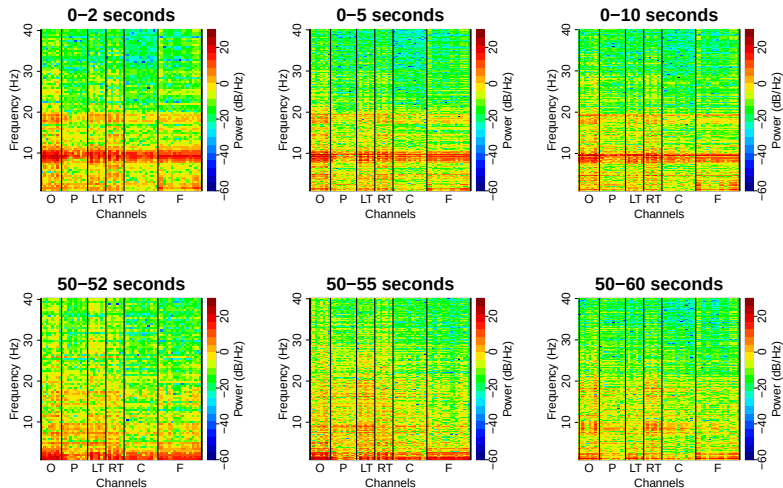


Figure 17: Estimated power spectral density at different channels and slices.

# Correlations between Channels for Subject 1

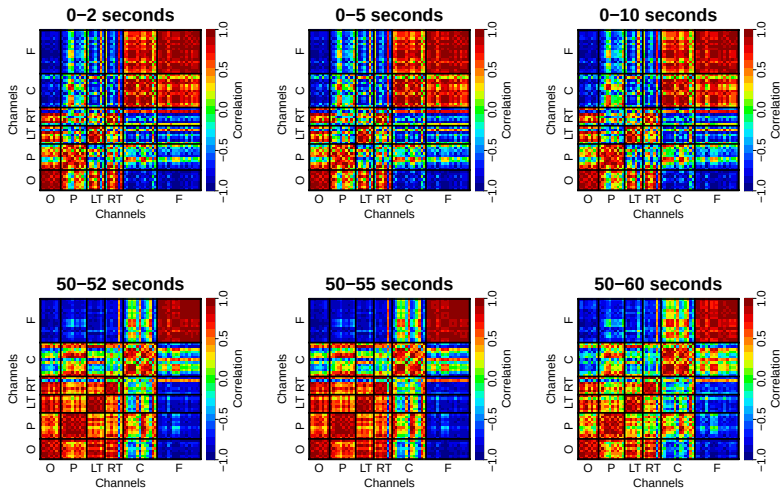


Figure 18: Estimated channel-by-channel correlation matrix at different slices.

# Multi-Subject Example

# Simultaneous Components across Slices and Subjects

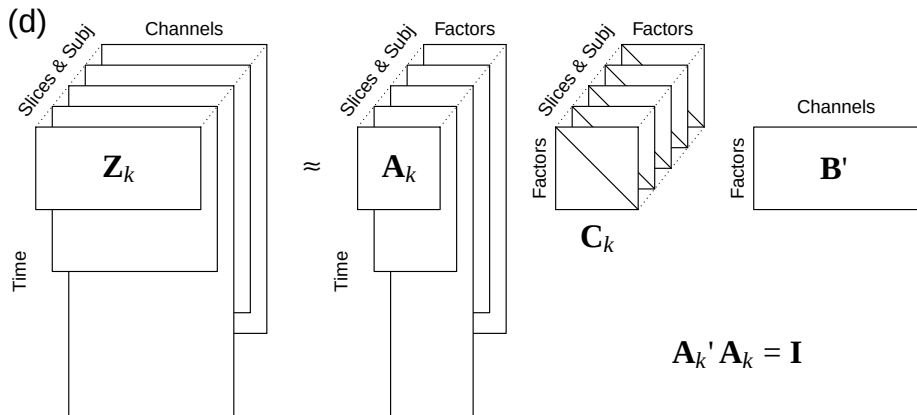


Figure 19: Parafac2 with orthogonal Mode A weights (SCA-IND).

# Choosing Number of Factors

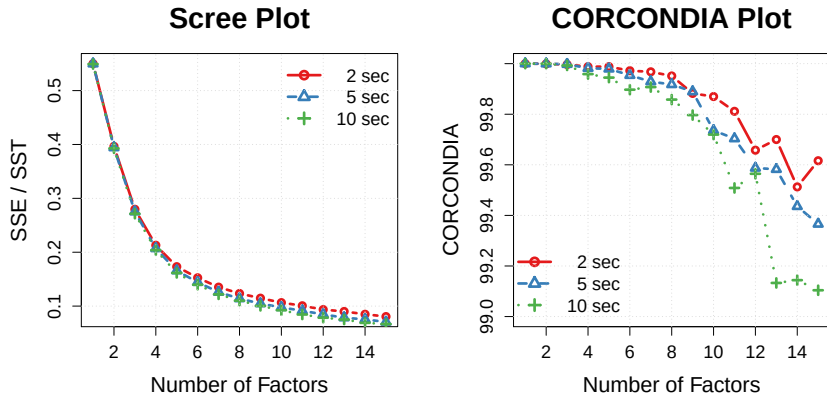


Figure 20: Scree and core consistency diagnostic (CORCONDIA) plots.

# Factor Loadings for Mode B for Subject 1

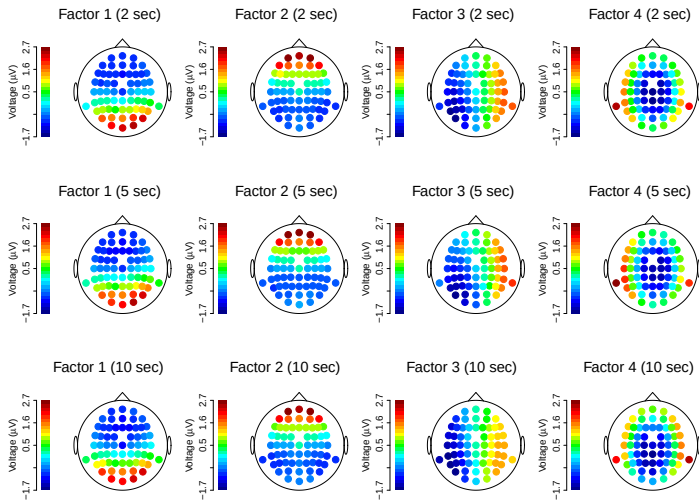


Figure 21: Estimated loadings for Mode B (channels) common to all subjects.

# Power Spectral Density of Factor Scores for Subject 1

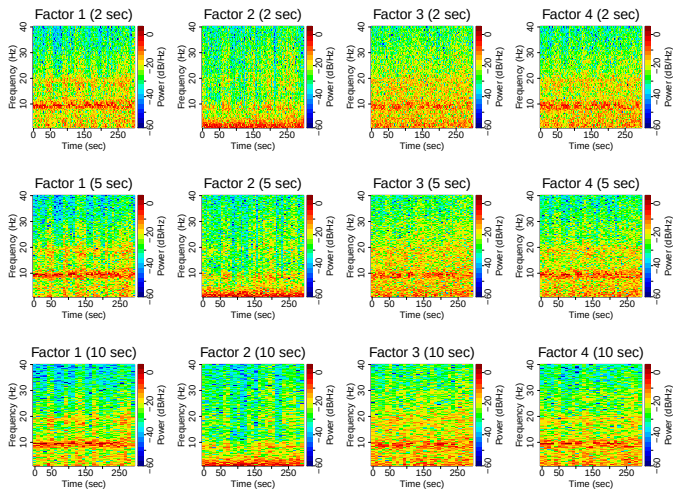


Figure 22: Estimated power spectral density of subject 1's Mode A (time) weights.

# Factor Loadings for Mode C for Subject 1

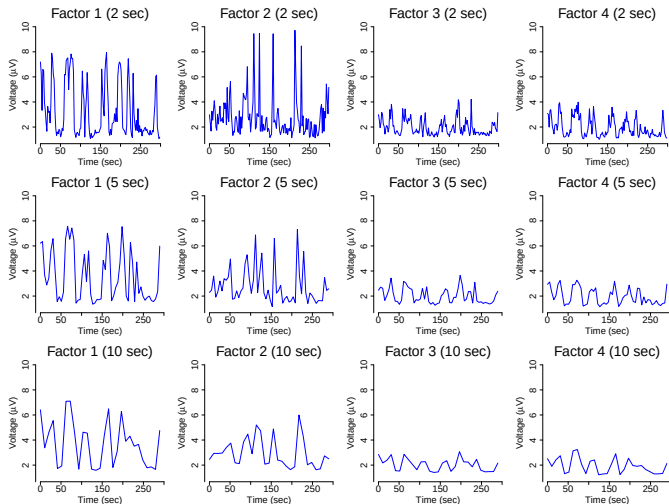


Figure 23: Subject 1's estimated loadings for Mode C (slices).

# Power Spectral Density of Data for Subject 1

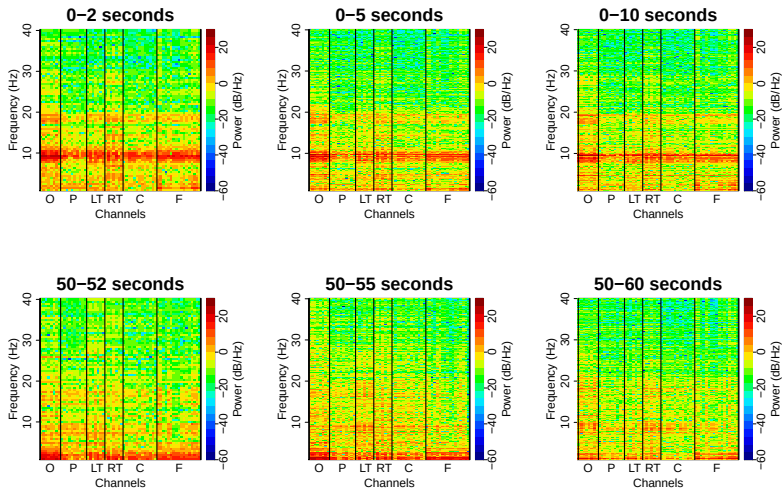


Figure 24: Subject 1's power spectral density at different channels and slices.

# Correlations between Channels for Subject 1

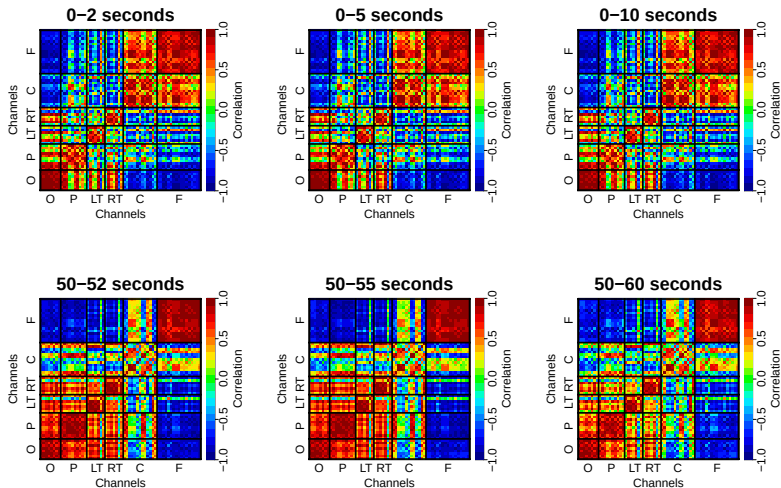


Figure 25: Subject 1's channel-by-channel correlation matrix at different slices.

# Conclusions and Extensions

## Conclusions and Findings:

- Genuine components  $\iff$  local and global variation
- Parafac2 (SCA) has great potential for resting state data
- rsEEG data show remarkable multilinear structure

## Extensions and Future Directions

- Resampling methods (bootstrap) for inference
- Connection to outcomes (e.g., regression and classification)
- Applications to other neuroimaging data (e.g., MEG or fMRI)

# References

- Cattell, R. B. (1944). “Parallel proportional profiles” and other principles for determining the choice of factors by rotation. *Psychometrika* 9, 267–283.
- Cattell, R. B. (1952). The three basis factor-analytic research designs — their interrelations and derivatives. *Psychological Bulletin* 49, 499–520.
- Harshman, R. A. (1970). Foundations of the PARAFAC procedure: Models and conditions for an “explanatory” multimodal factor analysis. *UCLA Working Papers in Phonetics* 16, 1–84.
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- Sockeel, S., D. Schwartz, M. Pélégriani-Issac, and H. Benali (2016). Large-scale functional networks identified from resting-state EEG using spatial ICA. *PLoS ONE* 11(1), e0146845.
- Timmerman, M. E. and H. A. L. Kiers (2003). Four simultaneous component analysis models for the analysis of multivariate time series from more than one subject to model intraindividual and interindividual differences. *Psychometrika* 68, 105–121.